8. Bohm actually rediscovered and further developed an approach that goes back to Prince Louis de Broglie, so this approach is sometimes called the de Broglie–Bohm approach.

9. For the mathematically inclined reader, Bohm's approach is local in *configuration* space but certainly *nonlocal* in real space. Changes to the wavefunction in one location in real space immediately exert an influence on particles located in other, distant locations.

10. For an exceptionally clear treatment of the Ghirardi-Rimini-Weber approach and its relevance to understanding quantum entanglement, see J. S. Bell, "Are There Quantum Jumps?" in *Speakable and Unspeakable in Quantum Mechanics* (Cambridge, Eng.: Cambridge University Press, 1993).

11. Some physicists consider the questions on this list to be irrelevant by-products of earlier confusions regarding quantum mechanics. The wavefunction, this view professes, is merely a theoretical tool for making (probabilistic) predictions and should not be accorded any but mathematical reality (a view sometimes called the "Shut up and calculate" approach, since it encourages one to use quantum mechanics and wavefunctions to make predictions, without thinking hard about what the wavefunctions actually mean and do). A variation on this theme argues that wavefunctions never actually collapse, but that interactions with the environment make it *seem* as if they do. (We will discuss a version of this approach shortly.) I am sympathetic to these ideas and, in fact, strongly believe that the notion of wavefunction collapse will ultimately be dispensed with. But I don't find the former approach satisfying, as I am not ready to give up on understanding what happens in the world when we are "not looking," and the latter—while, in my view, the right direction—needs further mathematical development. The bottom line is that measurement causes something that *is* or is *akin to* or *masquerades as* wavefunction collapse. Either through a better understanding of environmental influence or through some other approach yet to be suggested, this apparent effect needs to be addressed, not simply dismissed.

12. There are other controversial issues associated with the Many Worlds interpretation that go beyond its obvious extravagance. For